

Acoustical Characteristics of Tinnitus

An Analysis

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The recognition of tinnitus aurium as a problem in otological practice and in more recent years as a problem in audiological practice is widespread. Unfortunately, there has been little published research on this symptom. It may be hoped that the report by Reed⁷ signaled the beginning of systematic investigations of nonvibratory tinnitus aurium. Clinical observations seem to indicate that studies of tinnitus by several medical and nonmedical specialties may be of value. Our training as audiologists dictated that our investigation focus on the acoustical characteristics of the symptom.

Since there were no published studies of the various acoustical characteristics of tinnitus when the present investigation was undertaken, 2 primary goals were set. The first objective was to develop a basic methodology for use in the measurement of the acoustical characteristics of nonvibratory tinnitus aurium. The other primary goal was to determine whether or not group differences existed between normal ears and ears with various types of hearing loss for the characteristics tested. Three of these characteristics studied will be discussed: differences in the pitch of tinnitus; differences in the loudness of tinnitus, and the occurrence of tinnitus in normal ears.

Procedure

Subjects.—One hundred individuals were tested who met the criteria set for inclusion in the study. The subjects for the study were divided into 4 groups on the basis of their pure-tone thresholds for air conduction and bone conduction. The 4 groups were defined as follows and contained the number of subjects indicated:

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(1) Sensorineural ($N=30$): To be included in this group an individual was required to have a medically significant hearing loss* by air-conduction and bone-conduction thresholds no more than 10 db. better than the air-conduction threshold at any test frequency.

(2) Mixed Sensorineural-Conductive ($N=25$): To be included in this group a medically significant hearing loss by both air conduction and bone conduction and an air-bone gap of at least 15 db. at 2 or more frequencies were required.

(3) Conductive ($N=20$): To be included in this group an individual had to have a medical significant loss by air conduction, bone-conduction thresholds within normal range, and an air-bone gap of at least 15 db. at 3 frequencies.

(4) Normal ($N=25$): To be included in the normal-hearing group an individual was required to have pure-tone air-conduction thresholds no worse than 15 db. at 4,000 cps or below and no worse than 20 db. at higher frequencies.

In addition to the audiometric requirements, the following criteria were used in the selection of subjects: (1) All individuals with hearing loss were required to have tinnitus at the time of testing. (2) It was required that normal-hearing subjects have no active aural pathology at the time of the tests.

Equipment.—With the exceptions of 1 subject in the mixed group and 1 subject in the conductive group, all subjects were tested in a 2-room test suite at the Stanford University Speech and Hearing Clinic. Measurements in the test room with a General Radio Sound Level Meter (Type 1551-B) indicated an ambient noise level of 36 db. on the "C" scale and less than 24 db. on the "A" scale. The audiometric equipment consisted of an Allison model 21-B audiometer, a Hewlett-Packard model 201-C audio oscillator, and a matched set of TDH-39 earphones in MX41AR cushions. A locally constructed input matching transformer was used to match the output of the audio oscillator to the input of the audiometer. The test equipment used for the 2 subjects not examined at Stanford had essentially the same characteristics as that described above.

Methodology.—Six tests were used to evaluate the acoustical characteristics of tinnitus in the 3 hard-of-hearing groups. These tests in their order of administration were as follows.

* A medically significant hearing loss was arbitrarily defined as a loss of 20 db. or more at any 3 of the following test frequencies: 125, 250, 500, 750, 1,000, 1,500, 2,000, 3,000, 4,000, 6,000, or 8,000 cps; or a loss of at least 30 db. at any of the frequencies tested below 6,000 cps.

Subjective Evaluation of Tinnitus: A written description of what the tinnitus sounded like and a judgment of its loudness were obtained for each subject. The loudness judgment was made by having the subject place a check mark along a line which had the words *very soft* at one end and *very loud* at the other. This test was conducted as the initial test even though it was not known at the time whether an individual would meet the audiometric requirements for the study. The test order was established so that, if an individual qualified for the study, his subjective evaluation would not be influenced by stimuli presented during the pure tone tests.

Pure Tone Threshold Tests: Air-conduction and bone-conduction thresholds for pure tones in each ear were obtained at all of the test frequencies available on the audiometer. The testing method outlined by Carhart and Jerger¹ was used in these assessments.

Loudness Judgment of 1,000 CPS Tone: A tone of 1,000 cps was presented at a sensation level of 20 db. relative to the better ear for a period of 15 seconds. In this and in all subsequent tests of the hard-of-hearing groups, stimuli were presented bilaterally, although in many cases they were heard monaurally. At the conclusion of the 15-second listening period, each subject was asked to judge the loudness of what he had heard. The response sheet for this test was identical to the one used in the first test when the tinnitus was the stimulus to be judged.

Initial Pitch Match of Tinnitus: A gross evaluation of the pitch characteristics of each subject's tinnitus was made using the 11 frequencies available on the audiometer. Each of the 11 frequencies was presented twice at a sensation level of 10 db. relative to the better ear. The subjects were instructed to indicate on a response sheet whether a given stimulus was higher, lower, or the same pitch as their tinnitus. This test delineated a frequency range for the final assessment of the pitch of the tinnitus.

Final Pitch Match of Tinnitus: After a 5-minute rest period, a more precise measurement of pitch was made. During the rest period, the preceding test results were analyzed. Three regions were defined in this analysis: one in which all tones were judged to be higher in pitch than the tinnitus, one in which all tones were judged to be of lower pitch, and a final region of responses that were not consistent or in which the tones were judged to be of the same pitch as the tinnitus. Through the use of an audio oscillator, this final area was reduced by a bracketing technique until a frequency range of less than $\pm 5\%$ of the central frequency of the range was found in which the subject consistently matched the pitch of his tinnitus. This bracketing technique was repeated until the subject made 3 consecutive matches in the same frequency range. No further attempts were made to match the quality of the tinnitus. It should be noted that, when subjects reported tinnitus of a "noisy" quality, they were instructed to make all judgments in terms of the most predominant pitch.

Loudness Balance Test: The perceived loudness of each subject's tinnitus was quantified by a loudness balance type of test. The constant stimulus for the equal loudness judgment was the subject's tinnitus. The variable stimulus used for comparison was a pure tone which was, in each case, the next lower frequency on the audiometer from the perceived pitch of the subject's tinnitus. A lower frequency was employed to reduce the possibility of the 2 stimuli interacting to produce masking. A bracketing technique was used to find a range of no more than 3 db. in which repeated judgments of equal loudness were made. The difference between the lower limit of this range and the thresholds of the variable stimulus was taken as the loudness of the tinnitus.

Five tests were administered to assess the occurrence of tinnitus in the normal-hearing subjects. The tests in the order of their administration were as follows:

Pure-Tone Threshold Test: Complete air-conduction audiograms were obtained for each ear of the normal-hearing subjects, using the same testing method that was used with the hard-of-hearing subjects. Bone-conduction thresholds were not assessed for this group.

Loudness Judgment of 1,000 CPS Tone: A 1,000 cps tone was presented bilaterally for 15 seconds at a sensation level of 10 db. relative to the better ear. Subjects were instructed to mark a rating scale to indicate their judgment of the loudness of the stimulus following the presentation. In addition to a loudness judgment, subjects were asked to describe what the stimulus sounded like.

Pitch Memory Test: The stimulus from the preceding test served as the standard for assessing the ability of subjects to remember the pitch characteristics of a sound. The test frequencies of the audiometer other than 1,000 cps were presented twice in random order as the comparison stimuli. Each of these stimuli was presented bilaterally at a sensation level of 10 db. relative to the better ear. Following each presentation, the subjects indicated whether the comparison stimulus was higher, lower, or the same pitch as the standard.

Loudness Judgment of Subjective Sounds: This test and the one following it were based on the subject's perception in listening for sound in the test room for 5 minutes. During this time no auditory stimuli were presented by the examiner. The subject was instructed to listen closely to anything he heard, and he was then seated in the center of the room without earphones. Several dummy loudspeakers were placed around the room to reduce the possibility of the subject's expecting sound to come from a specific point. The subject was instructed to indicate on a rating scale the loudness of any sounds he heard and to describe what they sounded like.

Pitch Match of Subjective Sounds: Upon completion of the preceding test, the subjects were instructed to put on the earphones. When the earphones were in place, an evaluation was made of the pitch of the

sounds that the subject had perceived in the listening period. The audiometer provided the comparison stimuli for this assessment. These stimuli were presented bilaterally at a sensation level of 5 db. relative to the better ear. The initial stimulus tone was 500 cps. Subsequent presentations were based on a subject's preceding response. If the initial judgment was that 500 cps was lower in pitch than the subjectively generated sound, the second presentation was made at 8,000 cps. On the other hand, if the subject reported that 500 cps was higher in pitch than the sound he perceived in the quiet room, the second presentation was at 125 cps. The range including the perceived pitch was then reduced until the subjective sound was localized within a half-octave.

Results

Significant or interesting information can be presented on 4 topics. In some cases these results are based on several tests, while in others they come from a single measure.

Subjective Description of Tinnitus.—The terms the hard-of-hearing subjects used to describe what their tinnitus sounded like were heterogeneous in nature and were not shown to be related to any other test conducted. The differences within subject groups were as marked as those between groups. The term "ringing" was the most common description used by each group. Among all 3 hard-of-hearing groups there were no significant differences in the means, medians, or standard deviations of the rating scale judgments of the loudness of tinnitus. Those in the normal-hearing group who experienced subjective sounds judged them to be significantly ($P < 0.01$) quieter than the other groups.

Pitch Measurements.—All subjects in the hard-of-hearing groups were able to match the pitch of their tinnitus to a pure tone. The "median" test⁶ was used to analyze these measurements in terms of group differences. The test showed no significant difference in the frequency characteristics of tinnitus between the sensorineural and mixed-hearing-loss groups. Comparisons in-

volving the conductive-hearing-loss group, however, showed significant differences from both the sensorineural ($P < 0.01$) and the mixed ($P < 0.05$) hearing-loss groups. The way in which this difference attained significance is felt to be of major importance. The sensorineural and mixed hearing-loss groups matched the pitch of their tinnitus to a wide frequency range from 155-7,800 cps for the sensorineural group and from 40-6,500 cps for the mixed group. The conductive hearing-loss group, on the other hand, matched the pitch of their tinnitus to a restricted frequency range from 120-1,400 cps.

Loudness Measurements.—The loudness balance technique described earlier showed the loudness of tinnitus for all hard-of-hearing groups to be essentially the same. Of the 73 hard-of-hearing subjects participating in this test (2 subjects were excluded because the pitch of their tinnitus was lower than the lowest frequency of the audiometer), and over half matched the loudness of their tinnitus to stimuli 5 db. or less above their thresholds.

Occurrence of Tinnitus in Normals.—Ten of the 25 normal-hearing subjects reported hearing some type of sound during the 5-minute listening period in the test room. During this period no auditory stimuli were presented by the examiner. The number of normal-hearing subjects who experienced tinnitus was too small to permit statistical comparisons of the acoustical characteristics of their tinnitus with the other groups. They seemed, however, to follow the distribution of the sensorineural and mixed hearing-loss groups in the measure frequency of their tinnitus.

Comment

The results of this study support some theoretical concepts concerning the physiological mechanism of tinnitus and tend to cast doubt on others. At 2 points the results have deviated sharply from previous investigations.

Pitch of Tinnitus and Type of Hearing Loss.—Support is offered to the theories of

Lothrop⁵ and Saltzman and Ersner⁸ by the finding that the tinnitus in conductive losses was of a low-frequency type. It has been stated by Fowler^{2,3} that tinnitus associated with sensorineural losses tends to coincide with sharp dips in the audiogram. In 23 cases in the sensorineural group, a difference in threshold of at least 30 db. was found within a one-octave range at some point in the audiogram. The frequency of the subjects' tinnitus was related to this difference in auditory threshold in only 9 cases. This finding thus does not support the theorized relationship.

Measured Frequency of Tinnitus.—The results of measurements made of the frequency of tinnitus in the present study may be compared to similar measurements by Reed⁷ of the central frequency of tinnitus. The χ^2 for "2 by K" tables,⁶ when applied to these data, is 50.24 with 4 degrees of freedom ($P < 0.001$). It was felt that this highly significant difference might be due, in part, to the larger percentage of cases with conductive hearing loss reported in the present investigation. To reduce the chance of bias in the sample, the 20 cases in the conductive-hearing-loss group were excluded and a second χ^2 was computed. This χ^2 equalled 26.69 with 4 degrees of freedom ($P < 0.001$). The difference remained highly significant. The only other apparent reason for such a difference is found in differences in methodology between the 2 studies. In Reed's study, the examiner used the patients' descriptions as a means of making an estimate of the frequency of the tinnitus. The initial presentations of the comparison stimuli were based on this estimate. It has already been noted that in the present study the descriptions of tinnitus were not found to be related to other measures. The danger in the examiner's estimating the frequency of the tinnitus is that, if the error is in presenting stimuli that are too high, the subject may make pitch matches to stimuli that are harmonics of his tinnitus. Twenty-two of the 75 hard-of-hearing subjects in the present study were found to have confusion in dealing

with harmonics of their tinnitus. If problems of harmonic confusion were present in a significant percentage of Reed's subjects, the differences in methodology may well account for the significant difference in the reported results of the 2 studies.

Occurrence of Tinnitus in Normal-Hearing Subjects.—Of the 25 normal-hearing subjects tested in the present study, 10 were found to experience tinnitus while in a heavily sound-treated room. Heller and Bergman⁴ have reported that 94% of a group of 80 subjects who stated that they had normal hearing experienced tinnitus in a similar situation. A χ^2 of 10.5 with 1 degree of freedom ($P < 0.01$) is found when the results of these 2 studies are compared. The only apparent difference in the procedures followed in the 2 studies is that Heller and Bergman did not assess the hearing of their subjects audiometrically †. Instead, they accepted a subject's statement that he had normal hearing. While accumulating 25 cases with audiometrically normal hearing for the present study, 11 individuals were tested who reported that they had normal hearing when, in fact, they did not meet the criteria set for normal hearing. Of these 11 individuals, 9 perceived sounds during the listening period. The results of this rejected group are much more compatible with those reported by Heller and Bergman than was the case with the audiometrically normal group.

Conclusions

The following conclusions are based on the results of the present study.

1. The descriptions made by subjects of the characteristics of their tinnitus do not necessarily relate to measurements of its acoustic properties.
2. The tinnitus experienced by persons with conductive hearing loss is composed of low-frequency sounds, suggesting that it may be produced by a different physiological mechanism than the tinnitus associated with other types of hearing loss.

† Personal communication from Dr. Heller, Jan. 12, 1960.

3. The loudness of a subject's tinnitus is not related to the audiometrically defined classification of his hearing loss.

4. The perception of tinnitus by normal-hearing individuals in an extremely quiet room is not as common as has been reported by previous investigators.

Summary

A methodology has been presented by which psychophysical measurements can be made of the pitch and loudness of tinnitus. Data are presented on the proposed tests for groups with sensorineural, mixed, and conductive hearing losses. The most significant difference between groups was found to be the low-frequency nature of tinnitus in the conductive-hearing-loss group. The occurrence of tinnitus in individuals with audiometrically normal hearing was investigated and found to be infrequent. The results of this investigation have been compared to those reported by other investigators and the differences have been discussed.

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